1.Compare theoretical (and often announced) values for transmission data rates with realistic data rates. What could be reasons for the sometimes big difference?

Theoretical (announced) values

- Based on ideal lab conditions
- Maximum modulation, coding, no interference
- Full bandwidth usage
- No competing users

Realistic data rates

- Shared medium (especially wireless)
- Interference (engines, lightning, other cells, WLANs)
- Distance from base station
- Obstacles, attenuation
- Protocol overhead
- Congestion
- Mobility and handovers

Reasons for big differences

- Wireless is a *shared medium*
- Signal fading, shadowing, multipath
- Regulatory spectrum limitations
- Control signaling overhead
- Cell load (many users share capacity)

2.How do fixed and wireless systems differ in this context?

Fixed Networks	**Wireless Networks**
Dedicated physical medium	Shared radio medium
Stable channel conditions	Variable channel conditions
Lower error rates	Higher loss rates
Higher predictable throughput	Throughput depends on mobility, interference
More secure (physical access needed)	Radio interface accessible to everyone

Wireless systems suffer from:

- Interference
- Jitter and higher delay
- Spectrum regulation limits
- Shared bandwidth

3.Compare advances in CPU power, displays, batteries, applications, data rates. What are limiting factors of today?

Advances

- **CPU power**: Massive improvements (multi-core, GHz range)
- **Displays**: Flat, lightweight, high resolution, low power
- **Applications**: From voice → multimedia → real Internet → IoT
- **Data rates**: GSM (kbit/s) → UMTS (Mbit/s) → LTE (>100 Mbit/s) → 5G (Gbit/s)

Batteries (Limiting Factor!)

- Battery technology improves slowly
- Power consumption still critical
- CPU power $\propto CV^2f$ (voltage & frequency matter)
- Wireless transmission consumes high energy

Main limiting factors today

- Energy/battery capacity
- Spectrum scarcity
- Interference
- Heat dissipation
- Security/privacy issues

4.What are pros and cons regarding push or pull services?

Push (system sends automatically)

Pros

- Immediate updates
- Useful for alerts (emergency, offers)

Cons

- Privacy concerns
- Spam risk
- Energy consumption
- Network load

Pull (user requests information)

Pros

- User-controlled
- Lower unnecessary traffic
- Better privacy

Cons

- Delay in receiving information
- Requires user action

5.The history of wireless and mobile communication is not only a history of successes, but also failures. Iridium, HiperLAN, WAP, WiMax are examples. Find reasons for their failure (or at least limited success)!

Iridium

- 66 satellites → extremely high cost
- Expensive devices
- GSM expanded rapidly → cheaper alternative
- Poor indoor reception
- Bankruptcy in 2000

HiperLAN

- European WLAN standard
- Complex
- Market already dominated by IEEE 802.11
- Lack of industry adoption

WAP

- Very limited user experience
- Small displays, slow networks
- Expensive data pricing
- Poor usability

WiMax

- Promoted as DSL replacement
- LTE became dominant
- Ecosystem fragmentation
- Limited operator adoption

Main reason: Market timing + ecosystem + cost + competition.

6.Check the current, newest standards for mobile communications (3gpp), WLAN (IEEE), PANs (Zigbee, Bluetooth)

Mobile (3GPP)

- 5G NR (New Radio)
- 5G Advanced ongoing
- Work towards 6G research

WLAN (IEEE 802)

- IEEE 802.11ax (Wi-Fi 6)
- 802.11be (Wi-Fi 7, emerging)

PAN

- Bluetooth 5.x / LE Audio
- Zigbee (IoT focus, low power)

Trend:

- Higher efficiency
- Lower latency
- Massive IoT support
- Energy optimization

7.Currently, network operators roll-out 5G technology (covered later in this course). Figure out the strategies of the different network operators while migrating towards 5G systems. Can you find reasons why a single common system is not in sight?

Operator Strategies

- Non-Standalone (NSA): 5G over LTE core
- Standalone (SA): Full 5G core
- Use of low-band (coverage)
- Mid-band (capacity)
- mmWave (high speed, small cells)

Why no single common system?

- Different spectrum allocations per country
- Different legacy infrastructures
- Political/economic interests
- Vendor ecosystems
- Cost constraints
- Rural vs urban requirements

8.4G or higher coverage is flattening way before reaching the world population. Give reasons!

- Rural areas economically unattractive
- High infrastructure costs
- Low return on investment
- Difficult terrain
- Energy supply issues
- Political instability
- Low income regions

9.Where do you use horizontal handover since many years?

Used for many years in:

- GSM cell-to-cell transitions
- UMTS cell changes
- LTE inter-cell handover
- Wi-Fi access point roaming

Example: Moving during a phone call between base stations.

10.What about vertical handovers? Give examples where they already exist!

Examples:

- Wi-Fi → LTE switch on smartphone
- VoWiFi → VoLTE
- Cellular → Satellite fallback
- Dual-SIM switching networks

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11. What are problems for vertical handovers?

- Different QoS mechanisms
- Different authentication methods
- IP address changes
- Session interruption
- Billing differences
- Security mechanisms differ
- Latency variations

Seamless mobility remains a global research goal.

12.The data rate of LTE is already higher than many DSL connections (> 100Mbit/s) – so why doing more research? Can't we just replace all DSL connections by LTE – problem solved?

Although LTE >100 Mbit/s:

Problems

- Shared medium (capacity per cell limited)
- Spectrum limitations
- Congestion in dense areas
- Higher latency variability
- Data caps

- Indoor coverage problems
- Backhaul limitations
- Energy efficiency issues

DSL/Fiber:

- Dedicated line
- Stable throughput
- No air interference

Therefore:
Mobile complements fixed networks — it does not replace them.